**Project 1: Bank Marketing — Data Exploration, Preparation and Majority-Class Baseline**

## 1. Introduction

The Bank Marketing dataset contains information about telephone marketing campaigns conducted by a Portuguese banking institution. The dataset includes demographic, financial, contact-related, campaign-related and economic variables. The dependent variable, y, indicates whether a customer subscribed to a term deposit.

The objective of this project is to develop a machine-learning model that predicts whether a customer will subscribe to a term deposit following a telephone marketing campaign by a Portuguese banking institution.

The target variable is y, with two possible outcomes:

- **yes:** The customer subscribed to a term deposit.
- **no:** The customer did not subscribe.

The model should help the bank identify customers who are more likely to subscribe allowing marketing resources to be directed more effectively.

## 2. Data exploration and short report

The bank-additional-full.csv dataset is the extended Bank Marketing dataset. In its standard form it contains 41,188 customer records, 20 input variables and one target variable.

**The variables describe several aspects of customers and marketing campaigns:**

| Variable group | Examples | Description |
|---|---|---|
| Demographic | age, job, marital, education | Customer characteristics |
| Financial | default, housing, loan | Credit and loan information |
| Contact-related | contact, month, day_of_week | How and when customers were contacted |
| Campaign-related | campaign, pdays, previous, poutcome | Current and previous marketing contacts |
| Economic indicators | emp.var.rate, cons.price.idx, euribor3m, nr.employed | Broader economic conditions |
| Target | y | Whether the customer subscribed |

**Target variable distribution**

In the standard dataset the target distribution is represented by;

| Outcome | Number of customers | Percentage |
|---|---|---|
| No subscription | 36,548 | 88.73% |
| Subscription | 4,640 | 11.27% |
| Total | 41,188 | 100% |

**These figures describe the standard, unmodified dataset.**

The dataset is imbalanced because most customers did not subscribe. This is important because a model could achieve high accuracy by predicting that every customer will not subscribe, while failing to identify potential subscribers.

**3. Data preparation and justification**

Initial exploration involves examining the dataset dimensions, variable types, descriptive statistics, missing values, unknown entries and duplicate records. The target variable is imbalanced, with approximately 88.73% of customers belonging to the non-subscription category and 11.27% belonging to the subscription category in the standard dataset.

Data preparation involves checking and removing exact duplicate records, investigating missing values and retaining unknown categorical entries as distinct categories. The target variable is separated from the independent variables. The duration variable is excluded because its value is only available after a telephone conversation has occurred. Including it would introduce data leakage if the model is intended to select customers before a campaign call.

The dataset is divided into training and testing subsets using an 80:20 ratio and stratified sampling. A majority-class baseline is then established using a classifier that always predicts the most frequent category, no. Although this baseline achieves approximately 88.73% accuracy on the standard dataset, it has zero recall for actual subscribers. Consequently, accuracy alone cannot adequately measure the usefulness of a predictive model for this business problem.

Future modelling should therefore consider class imbalance and prioritise measures such as recall, precision, F1-score, balanced accuracy and ROC-AUC. Precision-recall analysis will also be useful when assessing the model's ability to identify likely subscribers.